

Conjecture 9 of *Sets Reconstructible with Medial Axis* is true*

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Abstract

We prove Conjecture 9 of [1]: if $X \subset \mathbb{R}^n$ is closed, $p \in \text{conv } X$, and $d(p, X) > \lambda > 0$, then p is λ -reconstructible, i.e. p lies in an open ball $\mathbb{B}(a, d(a))$ centred at a point a of the λ -medial axis $M_\lambda(X)$ of Chazal–Lieutier [2]. The proof is elementary and variational: no flows, no semiconcavity, no compactness of X , and no machinery beyond nearest points, orthogonal projection onto a closed convex set, and the parallel-axis (variance) identity. The hypothesis is used through a single quantitative separation lemma, and the strict inequality $d(p, X) > \lambda$ is sharp. The statement in fact holds for all $p \in \overline{\text{conv}} X$ and degenerates for $\lambda = 0$ to an independent proof of the convex-hull part of Theorem 6 of [1].

1 Setting and statement

Throughout, $X \subset \mathbb{R}^n$ is a non-empty closed set, $d(a) := \text{dist}(a, X)$, and

$$m(a) := \{x \in X \mid \|a - x\| = d(a)\}$$

is the (non-empty, compact) set of points of X nearest to a . The medial axis is $M_X = \{a \in \mathbb{R}^n \mid \#m(a) > 1\}$. Following [1, 2], for $\lambda \geq 0$ the λ -medial axis $M_\lambda(X)$ is obtained from M_X by removing the points a whose nearest-point set fits in a closed ball of radius λ :

$$M_\lambda(X) = \{a \in M_X \mid \forall z \in \mathbb{R}^n : m(a) \not\subseteq \overline{\mathbb{B}}(z, \lambda)\}.$$

A point $p \in X^c$ is λ -reconstructible if $p \in \bigcup_{a \in M_\lambda(X)} \mathbb{B}(a, d(a))$ [1, Def. 5].

Theorem 1.1 (Conjecture 9 of [1], strengthened to the closed hull). *Let $X \subset \mathbb{R}^n$ be a non-empty closed set, let $\lambda > 0$, and let $p \in \overline{\text{conv}} X$ satisfy $d(p) > \lambda$. Then there exists $a \in M_\lambda(X)$ with $\|p - a\| < d(a)$; in particular p is λ -reconstructible.*

Note that $d(p) > \lambda > 0$ forces $p \notin X$, so the statement is well-posed. We fix from now on

$$d_0 := d(p) > \lambda, \quad \beta_0 := \sqrt{d_0^2 - \lambda^2} > 0.$$

Call a point $a \in \mathbb{R}^n$ *captured* if $\|a - p\| < d(a)$, i.e. if the open free ball $\mathbb{B}(a, d(a))$ contains p . Since $m(a)$ is a singleton for $a \notin M_X$ and a singleton fits in a closed λ -ball, producing a *captured*

*Companion note to our formalization notes on Theorem 6 of [1]. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand at the level of detail a proof assistant requires. Remaining errors are ours.

a whose $m(a)$ fits in no closed λ -ball immediately gives $a \in M_X$ and then $a \in M_\lambda(X)$, proving the theorem. We therefore argue by contradiction and assume:

$$\text{for every captured } a \text{ there exists } z_a \in \mathbb{R}^n \text{ with } m(a) \subseteq \overline{\mathbb{B}}(z_a, \lambda). \quad (\text{H})$$

The whole proof is organized around the *power function*

$$\psi(a) := d(a)^2 - \|a - p\|^2,$$

the (negated) power of the point p with respect to the sphere $\partial\mathbb{B}(a, d(a))$. Thus a is captured iff $\psi(a) > 0$, and $\psi(p) = d_0^2$.

2 Two lemmas

Lemma 2.1 (separation). *Let $S \subseteq \mathbb{R}^n$ be non-empty with $S \subseteq \overline{\mathbb{B}}(z, r)$ for some $z \in \mathbb{R}^n$, $r \geq 0$, and let $q \in \mathbb{R}^n$ satisfy $\text{dist}(q, S) \geq D > r$. Then:*

- (i) *every $y \in \text{conv } S$ satisfies $\|q - y\| \geq \sqrt{D^2 - r^2} > 0$;*
- (ii) *there is a unit vector u with $\langle u, q - x \rangle \geq \sqrt{D^2 - r^2}$ for all $x \in S$.*

Proof. (i) Let $y = \sum_i \theta_i x_i$ be a finite convex combination of points $x_i \in S$. The parallel-axis identity (expand and use $\sum_i \theta_i (y - x_i) = 0$) gives, for any $w \in \mathbb{R}^n$,

$$\sum_i \theta_i \|w - x_i\|^2 = \|w - y\|^2 + \sum_i \theta_i \|y - x_i\|^2.$$

Applying it once with $w = q$ and once with $w = z$,

$$\|q - y\|^2 = \sum_i \theta_i \|q - x_i\|^2 - \sum_i \theta_i \|y - x_i\|^2 \geq \sum_i \theta_i \|q - x_i\|^2 - \sum_i \theta_i \|z - x_i\|^2 \geq D^2 - r^2.$$

(ii) Let $C := \overline{\text{conv } S}$, a non-empty closed convex set; by continuity, (i) holds for all $y \in C$. Let π be the orthogonal projection of q onto C and $\beta := \|q - \pi\|$; by (i), $\beta \geq \sqrt{D^2 - r^2} > 0$, so $u := (q - \pi)/\beta$ is well defined. The obtuse-angle characterization of the projection gives $\langle q - \pi, x - \pi \rangle \leq 0$ for every $x \in C$, hence for $x \in S$

$$\langle u, q - x \rangle = \langle u, q - \pi \rangle + \langle u, \pi - x \rangle = \beta - \frac{1}{\beta} \langle q - \pi, x - \pi \rangle \geq \beta \geq \sqrt{D^2 - r^2}. \quad \square$$

Lemma 2.2 (ascent step). *Assume (H) and let a be captured. Then there is a unit vector u such that for every $\varepsilon > 0$ there is $h_0 > 0$ with*

$$\psi(a + hu) \geq \psi(a) + 2h(\beta_0 - \varepsilon) \quad \text{for all } h \in (0, h_0).$$

Proof. Since $m(a) \subseteq X$ we have $\text{dist}(p, m(a)) \geq d_0$, and (H) provides $m(a) \subseteq \overline{\mathbb{B}}(z_a, \lambda)$ with $\lambda < d_0$. Lemma 2.1 (with $S = m(a)$, $q = p$, $r = \lambda$, $D = d_0$) yields a unit vector u with

$$\langle u, p - x \rangle \geq \beta_0 \quad \text{for all } x \in m(a). \quad (1)$$

Exact increment. Fix $h > 0$ and pick any $y \in m(a + hu)$. Then

$$d(a + hu)^2 = \|a + hu - y\|^2 = \|a - y\|^2 + 2h \langle u, a - y \rangle + h^2, \quad \|a + hu - p\|^2 = \|a - p\|^2 + 2h \langle u, a - p \rangle + h^2,$$

so the h^2 terms cancel in the difference:

$$\psi(a + hu) = \|a - y\|^2 - \|a - p\|^2 + 2h \langle u, p - y \rangle \geq \psi(a) + 2h \langle u, p - y \rangle, \quad (2)$$

using $\|a - y\| \geq d(a)$ (valid because $y \in X$). Note that (2) is exact: there is no second-order error term.

The new nearest point still sees the gap. We claim that for every $\varepsilon > 0$ there is $h_0 > 0$ such that every $h \in (0, h_0)$ and every $y \in m(a + hu)$ satisfy $\langle u, p - y \rangle \geq \beta_0 - \varepsilon$; combined with (2) this proves the lemma. Suppose not: there are $\varepsilon_0 > 0$, $h_k \downarrow 0$ and $y_k \in m(a + h_k u)$ with $\langle u, p - y_k \rangle < \beta_0 - \varepsilon_0$. Since d is 1-Lipschitz,

$$\|y_k - a\| \leq \|y_k - (a + h_k u)\| + h_k = d(a + h_k u) + h_k \leq d(a) + 2h_k,$$

so (y_k) is bounded; passing to a subsequence, $y_k \rightarrow y^* \in X$ (X is closed). Then $\|a - y^*\| = \lim_k \|a - y_k\| \leq d(a)$, while $\|a - y^*\| \geq d(a)$ because $y^* \in X$; hence $y^* \in m(a)$ and $\langle u, p - y^* \rangle \geq \beta_0$ by (1). But $\langle u, p - y^* \rangle = \lim_k \langle u, p - y_k \rangle \leq \beta_0 - \varepsilon_0$, a contradiction. $\square$

Remark 2.3. Lemma 2.2 is the engine of the proof, and it is worth recording the heuristics. Under (H) the nearest points of any captured a are clustered in a λ -ball, while all of them are at distance $\geq d_0 > \lambda$ from p ; Lemma 2.1 converts this into a *uniform* transversality gap (1), with the same constant β_0 for every captured a , no matter how large $d(a)$ is. Moving the centre in the direction u may well *decrease* the radius d and the depth $d - \|\cdot - p\|$; what it always increases, at rate $2\beta_0$, is the power ψ . This is what the strict hypothesis $d_0 > \lambda$ buys; when $d_0 = \lambda$, $\beta_0 = 0$ and the engine stalls (cf. Remark 4.1).

3 Proof of Theorem 1.1

Assume (H). Set

$$\kappa := \frac{\beta_0}{2}, \quad \varepsilon := \frac{\beta_0}{4}, \quad \Phi(a) := \psi(a) - 2\kappa \|a - p\|, \quad \mathcal{G} := \{a \in \mathbb{R}^n \mid \Phi(a) \geq d_0^2\}.$$

Φ is continuous (d is 1-Lipschitz), and $\Phi(p) = \psi(p) = d_0^2$, so $p \in \mathcal{G}$ and $\mathcal{G}$ is non-empty and closed. Moreover

$$a \in \mathcal{G} \implies \psi(a) \geq d_0^2 + 2\kappa \|a - p\| > 0, \quad (3)$$

so every point of $\mathcal{G}$ is captured.

Step 1: $\sup_{a \in \mathcal{G}} d(a) = \infty$. Suppose instead $\bar{D} := \sup_{\mathcal{G}} d < \infty$. For $a \in \mathcal{G}$, (3) gives $2\kappa \|a - p\| \leq \psi(a) - d_0^2 \leq d(a)^2 - d_0^2 \leq \bar{D}^2 - d_0^2$, so $\mathcal{G}$ is bounded, hence compact. The continuous function Φ attains its maximum over $\mathcal{G}$ at some $a^* \in \mathcal{G}$. By (3), a^* is captured, so Lemma 2.2 (with $\varepsilon = \beta_0/4$) supplies a unit u and an $h > 0$ with $\psi(a^* + hu) \geq \psi(a^*) + 2h \cdot \frac{3\beta_0}{4}$. Since $\|a^* + hu - p\| \leq \|a^* - p\| + h$,

$$\Phi(a^* + hu) \geq \Phi(a^*) + 2h \left(\frac{3\beta_0}{4} - \kappa \right) = \Phi(a^*) + \frac{\beta_0}{2} h > \Phi(a^*) \geq d_0^2.$$

Thus $a^* + hu \in \mathcal{G}$ and $\Phi(a^* + hu) > \Phi(a^*)$, contradicting maximality. Hence $\sup_{\mathcal{G}} d = \infty$.

Step 2: a limiting half-space. By Step 1 choose $a_k \in \mathcal{G}$ with $d_k := d(a_k) \rightarrow \infty$, and put $s_k := \|a_k - p\|$, $\psi_k := d_k^2 - s_k^2$. Since d is 1-Lipschitz, $d_k \leq d_0 + s_k$, so $s_k \geq d_k - d_0 \rightarrow \infty$; discard

the finitely many k with $s_k = 0$ and set $w_k := (a_k - p)/s_k$. By (3), $\psi_k \geq d_0^2 + 2\kappa s_k$. For every $y \in X$ we have $\|y - a_k\| \geq d_k$, i.e. $\|(y - p) - s_k w_k\|^2 \geq d_k^2$; expanding,

$$\langle w_k, y - p \rangle \leq \frac{\|y - p\|^2 + s_k^2 - d_k^2}{2s_k} = \frac{\|y - p\|^2 - \psi_k}{2s_k} \leq \frac{\|y - p\|^2 - d_0^2}{2s_k} - \kappa.$$

Passing to a subsequence with $w_k \rightarrow w$ (unit sphere compactness) and letting $k \rightarrow \infty$ with $y \in X$ fixed:

$$\langle w, y - p \rangle \leq -\kappa \quad \text{for every } y \in X. \quad (4)$$

Step 3: contradiction with $p \in \overline{\text{conv}} X$. Inequality (4) is linear in y , hence it is inherited by every finite convex combination of points of X , i.e. by every $q \in \text{conv } X$, and by continuity by every $q \in \overline{\text{conv}} X$. Taking $q = p$ gives $0 = \langle w, p - p \rangle \leq -\kappa < 0$, absurd.

Therefore (H) is false: there exists a captured a such that $m(a)$ fits in no closed ball of radius λ . As observed in Section 1, such an a lies in M_X (a singleton $m(a)$ would fit in a λ -ball), hence in $M_\lambda(X)$, and $p \in \mathbb{B}(a, d(a))$. $\square$

4 Remarks

Remark 4.1 (sharpness). The strict inequality $d(p) > \lambda$ cannot be weakened to $d(p) \geq \lambda$. Take $X = \{-\lambda e_1, \lambda e_1\} \subset \mathbb{R}^n$ and $p = 0 \in \text{conv } X$, so $d(p) = \lambda$. Here $M_X = \{x_1 = 0\}$ and every $a \in M_X$ has $m(a) = X \subseteq \overline{\mathbb{B}}(0, \lambda)$, so $M_\lambda(X) = \emptyset$ and no point is λ -reconstructible.

Remark 4.2 (the case $\lambda = 0$). The proof used $\lambda > 0$ nowhere except through $\beta_0 = \sqrt{d_0^2 - \lambda^2}$, and it runs verbatim for $\lambda = 0$ (balls of radius 0 being points, (H) then reads “every captured point has a unique nearest point”). The conclusion is: every $p \in \overline{\text{conv}} X$ with $d(p) > 0$ lies in $\mathbb{B}(a, d(a))$ for some $a \in M_X$. This is precisely the convex-hull part of Theorem 6 of [1] (Proposition 2 and Corollary 5 there), re-proved without the central-set machinery: no maximal balls, no C_X , and no Motzkin–Fremlin inclusion $C_X \subseteq \overline{M_X}$ are needed. The argument lands directly in M_X .

Remark 4.3 (what the proof does *not* use). There is no smallest-enclosing-ball (circumradius) theory: hypothesis (H) is consumed in the raw form “ $m(a)$ fits in *some* closed λ -ball”. There is no distance flow, no semiconcavity or Clarke subdifferential, no upper semicontinuity of the multimap m beyond the one-point compactness argument inside Lemma 2.2, and no Euler polygons: the tilted functional $\Phi = \psi - 2\kappa \|\cdot - p\|$ replaces the whole trajectory construction by a single compact maximization (Step 1). Boundedness or convexity of X is never assumed.

Remark 4.4 (why the naïve attacks see a threshold). It is instructive to compare with the steepest-ascent attack along Lieutier’s flow [3]: moving a captured centre a away from (the projection of a onto) $\text{conv } m(a)$ grows d at rate $g = \sqrt{1 - \mathcal{F}^2/d^2} \leq 1$ (with $\mathcal{F} \leq \lambda$ the cluster radius under (H)), while the depth $d(a) - \|a - p\|$ can leak at rate $1 - g \approx \lambda^2/2d^2$. Integrating the leak from $d = d_0$ to ∞ gives a total loss $\frac{\lambda}{4} \log \frac{d_0 + \lambda}{d_0 - \lambda}$, which stays below the initial depth d_0 only for $d_0 \gtrsim 1.033 \lambda$ (the root of $x = \frac{1}{4} \log \frac{x+1}{x-1}$). Constrained-maximization variants of that idea stall similarly (at thresholds such as $d_0 \geq \frac{2}{\sqrt{3}} \lambda$). The two changes that remove the threshold are: (a) the step direction is aimed by Lemma 2.1 at the separation between p and the cluster $\text{conv } m(a)$, rather than by the gradient of d ; and (b) progress is measured by the power ψ , whose increment (2) is exact — the gain rate $2\beta_0$ is independent of $d(a)$, so nothing has to be integrated and nothing leaks.

Remark 4.5 (formalization). Every ingredient is mathlib-friendly: nearest points on a closed set in a proper space; 1-Lipschitzness of `Metric.infDist`; the parallel-axis identity (inner-product algebra over finite convex combinations); orthogonal projection onto a non-empty closed convex set and its obtuse-angle characterization (`Mathlib.Analysis.InnerProductSpace.Projection`); compactness of closed bounded sets and of the unit sphere; and an extreme-value argument. The contradiction scheme mirrors the one already used in our formalization of Theorem 6, and Lemma 2.2 is structurally the same one-point limit argument as the inflation dichotomy there.

References

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